

CS224W Homework 3

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1 GraphRNN (20 points)

Graph generation models produce entire graphs for tasks such as predicting graph evolution, generating new graph instances, and detecting anomalous graphs. GraphRNN generates a graph by sequentially adding nodes and edges. Given a graph G and a node ordering π , it maps the graph to a unique sequence of node- and edge-addition decisions S^π .

When a new node is added, GraphRNN predicts whether it connects to each existing node. For example, the node-level step for the fourth node consists of the three binary decisions

$$S_{4,1}^\pi, \quad S_{4,2}^\pi, \quad S_{4,3}^\pi.$$

1.1 BFS Generation Sequence (12 points)

Consider the grid graph shown in Question 1.1 of the official homework PDF. GraphRNN adds its nodes in breadth-first-search order, starting at node A . When exploring a node's neighbors, visit them in alphabetical order. For each new node, report a 1 for every edge that should be added to an existing node and a 0 otherwise.

Give the node order followed by the edge-level predictions for every node.

Solution. Node order. Starting from A , the alphabetically ordered neighbors of A are B, D . Processing B next discovers C, E , while processing D discovers no new node. Finally, processing C discovers F . Therefore the BFS ordering is

$$\pi = (A, B, D, C, E, F).$$

Edge-level predictions. For each new node, the entries below are listed against the preceding nodes in the order π :

new node	preceding nodes	predictions
A	none	none
B	A	(1)
D	A, B	(1, 0)
C	A, B, D	(0, 1, 0)
E	A, B, D, C	(0, 1, 1, 0)
F	A, B, D, C, E	(0, 0, 0, 1, 1)

Here 1 denotes an edge to the corresponding preceding node and 0 denotes no edge.

1.2 Advantages of BFS Ordering (8 points)

Explain two advantages of using a BFS node ordering for graph generation instead of a random node ordering. You may use claims from the original GraphRNN paper.

Solution.

1. BFS greatly reduces the number of redundant sequential representations of the same graph.

Let G have n nodes and let Π be the set of its $n!$ node permutations. With an unrestricted ordering, every $\pi \in \Pi$ can produce a different training sequence

$$S^\pi = f_S(G, \pi).$$

These sequences describe the same unlabeled graph, but an autoregressive model sees them as different ordered targets. It would therefore have to spread probability mass over, and learn the dependencies in, potentially $n!$ different representations of one graph.

GraphRNN instead applies a deterministic BFS procedure after choosing a starting node and a tie-breaking order:

$$b_G(\pi) = \text{BFS}(G, \pi), \quad S^\pi = f_S(G, b_G(\pi)).$$

The map b_G is many-to-one. For example, changing the relative positions in π of vertices that BFS does not compare can leave the starting node, every neighbor-discovery order, and hence the complete BFS output unchanged. Thus, if

$$\mathcal{B}(G) = \{ \text{BFS}(G, \pi) : \pi \in \Pi \},$$

then

$$|\mathcal{B}(G)| \leq |\Pi| = n!,$$

and the inequality is strict for many graphs. The model only needs to learn the sequences induced by $\mathcal{B}(G)$, rather than those induced by every arbitrary permutation.

This restriction does not remove any graph from the model's support. Every graph has at least one BFS traversal, and the full edge decisions associated with any such traversal reconstruct exactly the original graph. BFS therefore removes only redundant orderings, not graph structures. Fewer equivalent targets mean less ordering-induced variation, less probability mass split among artificial representations, and an easier sequence-learning problem. The reduction need not be strict in the worst case—a star graph can still have $n!$ BFS orderings—but it is substantial for many graph families.

2. BFS localizes possible edges and shortens every edge-generation sequence.

Under an arbitrary ordering $v_1, \dots, v_n$, the adjacency vector for the newly added node v_i is

$$S_i = (A_{1,i}, A_{2,i}, \dots, A_{i-1,i})^\top.$$

It has $i-1$ entries because, without additional structure, v_i might be adjacent to any preceding node. Consequently, generating an n -node graph requires

$$\sum_{i=2}^n (i-1) = \frac{n(n-1)}{2} = \Theta(n^2)$$

binary edge predictions.

BFS gives a stronger locality property. Consider the instant at which a new vertex v_i is first discovered. Let $\mathcal{F}_i$ consist of the vertex currently being expanded together with the already discovered but not yet expanded vertices in the BFS queue. We claim that every preceding neighbor of v_i belongs to $\mathcal{F}_i$.

To prove the claim, suppose instead that v_i were adjacent to a preceding vertex $u \notin \mathcal{F}_i$. Such a vertex u was fully processed before v_i was discovered. BFS examines every edge incident to u while processing u . It would therefore have discovered and enqueued v_i at that earlier time, contradicting the assumption that v_i is first discovered now. Hence old, completed parts of the BFS traversal cannot contain a neighbor of the new node; all possible preceding neighbors lie in the current frontier.

Because BFS processes nodes in FIFO discovery order, this frontier is a suffix of the preceding node sequence. Let

$$M = \max_i |\mathcal{F}_i|$$

be its maximum size. It is therefore sufficient to retain at most the last M entries of each adjacency vector:

$$S_i = (A_{\max(1, i-M), i}, \dots, A_{i-1, i})^\top,$$

with padding when $i - 1 < M$. All omitted entries are guaranteed to be zero by the BFS argument above. The total number of predictions is now bounded by

$$\sum_{i=2}^n \min(M, i-1) \leq M(n-1) = O(Mn).$$

The GraphRNN analysis relates M to the largest BFS layer:

$$M = O\left(\max_{1 \leq d \leq \text{diam}(G)} |\{v : \text{dist}(v, v_1) = d\}|\right).$$

Thus, whenever the BFS frontier is much smaller than the whole graph ($M \ll n$), generation is sub-quadratic. For example, a path has $M = O(1)$, giving $O(n)$ predictions, while a roughly square grid has $M = O(\sqrt{n})$, giving $O(n^{3/2})$. Dense graphs or stars may have $M = \Theta(n)$, so the worst case remains quadratic.

Besides reducing computation and memory, the fixed and shorter edge-level sequences remove many guaranteed-zero decisions and shorten the dependencies that the edge-level RNN must learn. BFS therefore makes both training and sampling more scalable than a random ordering, which can place adjacent vertices arbitrarily far apart.

2 Subgraphs and Order Embeddings (35 points)

For this question, graph A is a subgraph of graph B when some induced subgraph of B is graph-isomorphic to A . Anchor nodes are not considered, and every coordinate of an order embedding is non-negative.

The order-embedding constraint is

$$A \text{ is a subgraph of } B \iff z_A[i] \leq z_B[i] \text{ for every dimension } i.$$

Equivalently, write $z_A \preceq z_B$ for coordinate-wise inequality.

2.1 Transitivity (8 points)

Show that the subgraph relation is transitive: if graph A is a subgraph of graph B , and graph B is a subgraph of graph C , then graph A is a subgraph of graph C .

Use the subgraph-isomorphism definition: if A is a subgraph of B , then there is a bijection f from V_A to a subset of V_B such that the subgraph of B induced by

$$\{f(v) : v \in V_A\}$$

is graph-isomorphic to A .

Solution. Write the three graphs as

$$A = (V_A, E_A), \quad B = (V_B, E_B), \quad C = (V_C, E_C).$$

Because A is a subgraph of B , there is a subset $U_B \subseteq V_B$ and a bijection

$$f : V_A \longrightarrow U_B$$

such that f is a graph isomorphism from A to the induced subgraph $B[U_B]$. In particular, for every $u, v \in V_A$,

$$\{u, v\} \in E_A \iff \{f(u), f(v)\} \in E_B. \quad (1)$$

Similarly, because B is a subgraph of C , there are a subset $U_C \subseteq V_C$ and a bijection

$$g : V_B \longrightarrow U_C$$

such that g is a graph isomorphism from B to the induced subgraph $C[U_C]$. Thus, for every $x, y \in V_B$,

$$\{x, y\} \in E_B \iff \{g(x), g(y)\} \in E_C. \quad (2)$$

The right-hand side of (2) is exactly the edge relation in $C[U_C]$, because that subgraph is induced. Now define

$$U'_C = g(U_B) \subseteq U_C \quad \text{and} \quad h = g \circ f : V_A \longrightarrow U'_C.$$

Both f and g are bijections on their stated domains, so h is a bijection from V_A onto U'_C . For any $u, v \in V_A$, combining (1) and (2) gives

$$\begin{aligned} \{u, v\} \in E_A &\iff \{f(u), f(v)\} \in E_B \\ &\iff \{g(f(u)), g(f(v))\} \in E_C \\ &\iff \{h(u), h(v)\} \in E(C[U'_C]). \end{aligned}$$

The last equivalence holds because $C[U'_C]$ is the induced subgraph on U'_C : it contains precisely the edges of C whose two endpoints lie in U'_C . Therefore h preserves both edges and non-edges and is a graph isomorphism from A to $C[U'_C]$. Since $C[U'_C]$ is an induced subgraph of C , the definition of the subgraph relation implies that A is a subgraph of C . Hence the subgraph relation is transitive.

2.2 Anti-symmetry (8 points)

Using the same subgraph-isomorphism definition, show that if graph A is a subgraph of graph B , and graph B is a subgraph of graph A , then A and B are graph-isomorphic.

Hint. What does the condition imply about the numbers of nodes in A and B , and why does that implication result in graph isomorphism?

Solution. Write

$$A = (V_A, E_A), \quad B = (V_B, E_B),$$

and assume, as in the homework setting, that the graphs are finite. Since A is a subgraph of B , there are a subset $U_B \subseteq V_B$ and a bijection

$$f : V_A \longrightarrow U_B$$

such that A is graph-isomorphic to the induced subgraph $B[U_B]$. Consequently,

$$|V_A| = |U_B| \leq |V_B|. \tag{1}$$

Conversely, since B is a subgraph of A , there are a subset $U_A \subseteq V_A$ and a bijection from V_B onto U_A . Hence

$$|V_B| = |U_A| \leq |V_A|. \tag{2}$$

Combining (1) and (2) gives

$$|V_A| = |V_B|.$$

It follows from $U_B \subseteq V_B$ and $|U_B| = |V_A| = |V_B|$ that

$$U_B = V_B.$$

Therefore the induced subgraph $B[U_B]$ is the whole graph B , not a proper subgraph.

The original bijection $f : V_A \rightarrow U_B$ is consequently a bijection $f : V_A \rightarrow V_B$. Since it is an isomorphism from A to $B[U_B] = B$, for every $u, v \in V_A$ it satisfies

$$\{u, v\} \in E_A \iff \{f(u), f(v)\} \in E_B.$$

Thus f preserves both edges and non-edges and is a graph isomorphism between A and B . Hence the subgraph relation is anti-symmetric up to graph isomorphism.

2.3 Common Subgraph (4 points)

Consider a two-dimensional order-embedding space. Graphs A and B have embeddings $\mathbf{z}_A$ and $\mathbf{z}_B$, and suppose the order-embedding constraint is preserved perfectly. Prove that graph X is a common subgraph of A and B if and only if

$$\mathbf{z}_X \preceq \min\{\mathbf{z}_A, \mathbf{z}_B\},$$

where the minimum is taken coordinate-wise.

Solution. Let

$$\min\{\mathbf{z}_A, \mathbf{z}_B\} = (\min\{z_A[0], z_B[0]\}, \min\{z_A[1], z_B[1]\}).$$

By definition, X is a common subgraph of A and B exactly when X is a subgraph of A and X is a subgraph of B .

($\implies$) Assume that X is a common subgraph of A and B . Perfect preservation of the order-embedding constraint gives

$$\mathbf{z}_X \preceq \mathbf{z}_A \quad \text{and} \quad \mathbf{z}_X \preceq \mathbf{z}_B.$$

Therefore, for each coordinate $i \in \{0, 1\}$,

$$z_X[i] \leq z_A[i] \quad \text{and} \quad z_X[i] \leq z_B[i].$$

The number $z_X[i]$ is consequently no larger than the smaller of $z_A[i]$ and $z_B[i]$:

$$z_X[i] \leq \min\{z_A[i], z_B[i]\}, \quad i = 0, 1.$$

These two coordinate inequalities are precisely

$$\mathbf{z}_X \preceq \min\{\mathbf{z}_A, \mathbf{z}_B\}.$$

($\impliedby$) Conversely, assume that

$$\mathbf{z}_X \preceq \min\{\mathbf{z}_A, \mathbf{z}_B\}.$$

For each $i \in \{0, 1\}$, the definition of the minimum gives

$$\min\{z_A[i], z_B[i]\} \leq z_A[i] \quad \text{and} \quad \min\{z_A[i], z_B[i]\} \leq z_B[i].$$

Combining these inequalities with the assumption yields

$$z_X[i] \leq z_A[i] \quad \text{and} \quad z_X[i] \leq z_B[i] \quad \text{for } i = 0, 1.$$

Hence

$$\mathbf{z}_X \preceq \mathbf{z}_A \quad \text{and} \quad \mathbf{z}_X \preceq \mathbf{z}_B.$$

Applying the reverse direction of the perfect order-embedding constraint, X is a subgraph of both A and B . Thus X is a common subgraph of A and B .

Both implications hold, so

$$X \text{ is a common subgraph of } A \text{ and } B \iff \mathbf{z}_X \preceq \min\{\mathbf{z}_A, \mathbf{z}_B\}.$$

2.4 Incomparable Graphs (5 points)

Suppose A , B , and C are pairwise non-isomorphic graphs, and none is a subgraph of another. They are embedded into a two-dimensional order-embedding space. Without loss of generality, assume

$$z_A[0] > z_B[0] > z_C[0].$$

What must this imply about the relationship among $z_A[1]$, $z_B[1]$, and $z_C[1]$ if the order-embedding constraint is satisfied perfectly?

Solution. The second coordinate must be ordered in the opposite direction:

$$z_A[1] < z_B[1] < z_C[1].$$

To see this, first compare A and B . We are given

$$z_B[0] < z_A[0],$$

so the first coordinate already satisfies the inequality needed for $z_B \preceq z_A$. If we also had $z_B[1] \leq z_A[1]$, then both coordinates would satisfy

$$z_B[0] \leq z_A[0] \quad \text{and} \quad z_B[1] \leq z_A[1].$$

This would give $z_B \preceq z_A$. By the perfectly preserved order-embedding constraint, B would then be a subgraph of A , contrary to the hypothesis. Therefore

$$z_B[1] > z_A[1]. \tag{1}$$

Apply the same argument to B and C . Since $z_C[0] < z_B[0]$, the assumption $z_C[1] \leq z_B[1]$ would imply $z_C \preceq z_B$, and hence that C is a subgraph of B . This is also forbidden, so

$$z_C[1] > z_B[1]. \tag{2}$$

Combining (1) and (2) yields

$$\boxed{z_A[1] < z_B[1] < z_C[1]}.$$

The inequalities must be strict: equality in either comparison would still make the vector with the smaller first coordinate coordinate-wise no larger than the other vector, which would incorrectly assert a subgraph relation. With this reverse ordering, every pair remains incomparable: the graph with the larger first coordinate cannot be below the other one in dimension 0, and the graph with the smaller first coordinate is above it in dimension 1.

2.5 Limitation of a Two-Dimensional Order-Embedding Space (10 points)

Suppose again that A , B , and C are pairwise non-isomorphic graphs, and none is a subgraph of another. Construct graphs X , Y , and Z , each of which is a subgraph of one or more graphs in $\{A, B, C\}$, so that their embeddings must satisfy

$$z_X \preceq z_Y \quad \text{and} \quad z_X \preceq z_Z.$$

You do not need to give embedding coordinates or draw the graphs. Specify only which of A , B , and C contain X , Y , and Z as subgraphs, and explain why the inequalities are guaranteed.

Use the following clarifications:

1. Without loss of generality, $z_A[0] > z_B[0] > z_C[0]$.
2. Each of X , Y , and Z may be a common subgraph of exactly one, two, or three graphs in $\{A, B, C\}$.
3. The construction must guarantee both inequalities whenever the order-embedding constraint is satisfied in two dimensions.
4. It is not necessary to exhibit concrete graph drawings. The requested subgraph relationships and their justification are sufficient.

Solution. We use the following containment pattern:

- X is a common subgraph of all three graphs A, B, C ;
- Y is a subgraph of exactly B , so Y is not a subgraph of A or C ;
- Z is also a subgraph of exactly B , so Z is not a subgraph of A or C .

The graphs Y and Z can be chosen to be different subgraphs of B . The important point is that their listed memberships are exact.

From the assumption in the question and the result of the preceding part, the three parent embeddings have the strict ordering

$$z_A[0] > z_B[0] > z_C[0] \quad \text{and} \quad z_A[1] < z_B[1] < z_C[1]. \quad (1)$$

Because X is a subgraph of every parent, the order-embedding constraint gives

$$z_X \preceq z_A, \quad z_X \preceq z_B, \quad z_X \preceq z_C.$$

Taking the strongest bound in each coordinate and using (1),

$$z_X[0] \leq z_C[0] \quad \text{and} \quad z_X[1] \leq z_A[1]. \quad (2)$$

Consider Y . Since Y is a subgraph of B ,

$$z_Y[0] \leq z_B[0], \quad z_Y[1] \leq z_B[1]. \quad (3)$$

Moreover, Y is not a subgraph of A . If $z_Y[1] \leq z_A[1]$, then (3) and (1) would also give $z_Y[0] \leq z_A[0]$, and hence $z_Y \preceq z_A$. Perfect preservation would then imply that Y is a subgraph of A , a contradiction. Therefore

$$z_Y[1] > z_A[1]. \quad (4)$$

Similarly, Y is not a subgraph of C . The second inequality in (3) and (1) already imply $z_Y[1] \leq z_B[1] < z_C[1]$. Thus the only way that $z_Y \not\preceq z_C$ can hold is

$$z_Y[0] > z_C[0]. \quad (5)$$

Combining (2), (4), and (5) gives

$$z_X[0] \leq z_C[0] < z_Y[0], \quad z_X[1] \leq z_A[1] < z_Y[1],$$

so

$$z_X \preceq z_Y. \quad (6)$$

Exactly the same argument applies to Z : from $z_Z \preceq z_B$, together with $z_Z \not\preceq z_A$ and $z_Z \not\preceq z_C$, we obtain

$$z_Z[1] > z_A[1] \quad \text{and} \quad z_Z[0] > z_C[0].$$

Together with (2), this yields

$$z_X[0] \leq z_C[0] < z_Z[0], \quad z_X[1] \leq z_A[1] < z_Z[1],$$

and hence

$$z_X \preceq z_Z. \quad (7)$$

Thus this containment pattern guarantees both required inequalities in every perfect two-dimensional order embedding. The coordinate argument forces the embedding to place X below both Y and Z , even though the construction only specifies their relationships with A, B, C . We can choose B to contain X, Y, Z as suitable induced subgraphs, while A and C omit Y and Z . We may additionally choose X not to be a subgraph of either Y or Z , illustrating the two-dimensional representation limitation.

Honor Code

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